Football clubs, communities and convergence

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Football Finance, Analytics: Past Present and Future Conference





15 February 2025

Introduction

- Economics: what determines growth?
- Primary concept: Diminishing Marginal Returns.
  - ↑ factor of production yields decreasing returns in output.
- Implies convergence:
  - Those doing more gain less from using more inputs.
  - Developing countries should catch up to developed.
- But: this assumes some common, global, production function.

  - Empirical work has generally not supported convergence (Barro, 1991).

Introduction

- When applied to single industry, more support for the theory (Rodrik, 2011).
  - But what about data to measure this?
- Football is an industry:
  - Basic production technology: jumpers for goalposts.
  - Fierce competition leads to rapid learning.
  - Readily available productivity measure: winning.

Introduction

- Football is an industry with incredible data:
  - Over a century of data (Reade, 2021).
  - Regional detail (Castells-Quintana and Gasquez, 2024).
- Krause and Szymanski (2019) look at global convergence:
  - International football, 32,000 matches since 1950.
  - Win percentages over World Cup cycles display convergence.
  - Though 'convergence clubs' still exist: France, Germany,...

Introduction

- What about **regional** development?
  - Development **within** a country like England.
  - Buraimo et al. (2017): Football + emergence of N/S divide.
  - 'Levelling up': Ongoing persistence of gap, policy relevance.
- We use domestic football in England to measure convergence:
  - Fewer trade barriers within a country.
  - Do poorer performing regions 'catch-up'?
  - Or do local economic factors constrain some areas?
  - And do 'convergence clubs' exist/persist?

Data

- All football matches in English league and cup since 1871:
  - 416,819 matches up to February 10.
  - Date, teams, result and attendance.
- Sources: englishfootballleaguetables.co.uk and soccerbase.com
- Regional data:
  - Manual geo-coding of football stadiums for clubs.
  - Match clubs to local authority districts (LADs), regions.
  - LAD population data from Census and ONS. Data
  - LAD GDP data from ONS. Data

Method

- Need to measure productivity. Need output:

$$outcome_{ijt} = \begin{cases} 1 & \text{if home team } (i) \text{ win match at time } t, \\ 0.5 & \text{if draw is outcome of match at time } t, \\ 0 & \text{if away team } (j) \text{ win match at time } t. \end{cases} \tag{1}$$

- First step: do local economic conditions influence match outcomes?
  - If so, may rule out unconditional convergence.

Method

- Run linear regression to test:

$$outcome_{ijt} = \beta_0 + \beta_1 \frac{GDP_{it}}{GDP_{it} + GDP_{jt}} + \beta_2 \frac{pop_{it}}{pop_{it} + pop_{jt}} + \beta_3 \frac{exp_{it}}{exp_{it} + exp_{jt}} + \beta_4 pos_{it} + \beta_5 pos_{jt} + \beta_6 elo_{ijt} + \varepsilon_{it}. \quad (2)$$

- Exp=experience, total matches played by a team.

Elo ratings | Population | GDP per capita | Full position

Method

- Output: win percentage augmented for the draw:

$$y_{it} = \frac{1}{N} \sum_{j=1}^{N} outcome_{ijt}, \tag{3}$$

- Calculated at LAD level.
- Unconditional convergence is $\beta < 0$ in:

$$\Delta y_{it} = \alpha + \beta y_{i,t-1} + \varepsilon_{it}. \tag{4}$$

- Aggregate to annual, and five-year time periods.

Results – match outcomes

	Dependent variable:			
	outcome			
	(1)	(2)	(3)	(4)
Elo prediction			0.706***	0.756***
			(0.015)	(0.007)
GDP share	−0.113		−0.110	
	(0.078)		(0.077)	
Population share	0.270***	0.006	0.151	0.007
	(0.104)	(0.008)	(0.102)	(0.008)
Experience share	0.048***	0.041***	0.025***	0.010***
	(0.005)	(0.003)	(0.005)	(0.003)
Observations	91,490	344,499	91,490	344,499
R^2	0.058	0.038	0.080	0.068
Adjusted R^2	0.051	0.035	0.073	0.065
Residual Std. Error	0.416 (df = 90800)	0.412 (df = 343438)	0.412 (df = 90799)	0.405 (df = 343437)
Note:				*p<0.1; **p<0.05; ***p<0.01

Results – match outcomes league and cup

	Dependent variable:			
	outcome			
	(1)	(2)	(3)	(4)
	League		Cup	
Elo predict	0.765***	0.581***	0.701***	0.469***
	(0.008)	(0.011)	(0.022)	(0.033)
Full position (H)		−0.003***		−0.002***
		(0.0001)		(0.0002)
Full position (A)		0.002***		0.002***
		(0.0001)		(0.0002)
Population share	0.010	0.006	0.027	−0.041
	(0.009)	(0.009)	(0.026)	(0.028)
Experience share	0.009***	0.008***	0.042***	0.017*
	(0.003)	(0.003)	(0.009)	(0.010)
Observations	295,510	270,145	34,245	24,843
R^2	0.062	0.061	0.143	0.144
Adjusted R^2	0.060	0.059	0.117	0.121
Residual Std. Error	0.404 (df = 294774)	0.404 (df = 269461)	0.406 (df = 33215)	0.402 (df = 24201)
Note:				*p<0.1; **p<0.05; ***p<0.01

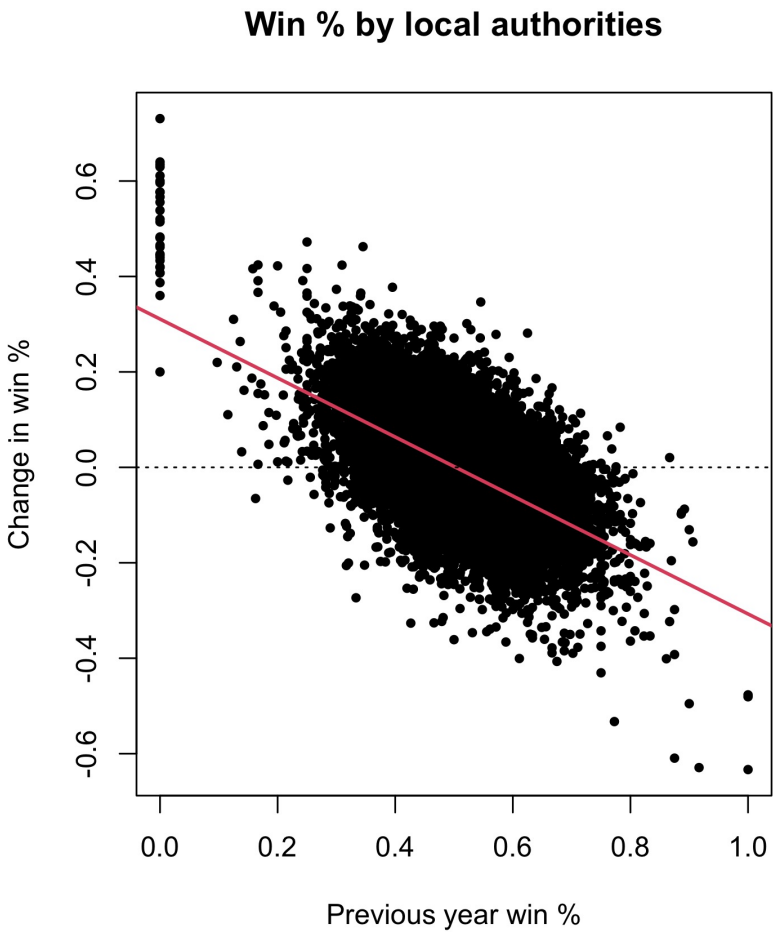
Results – match outcomes over time

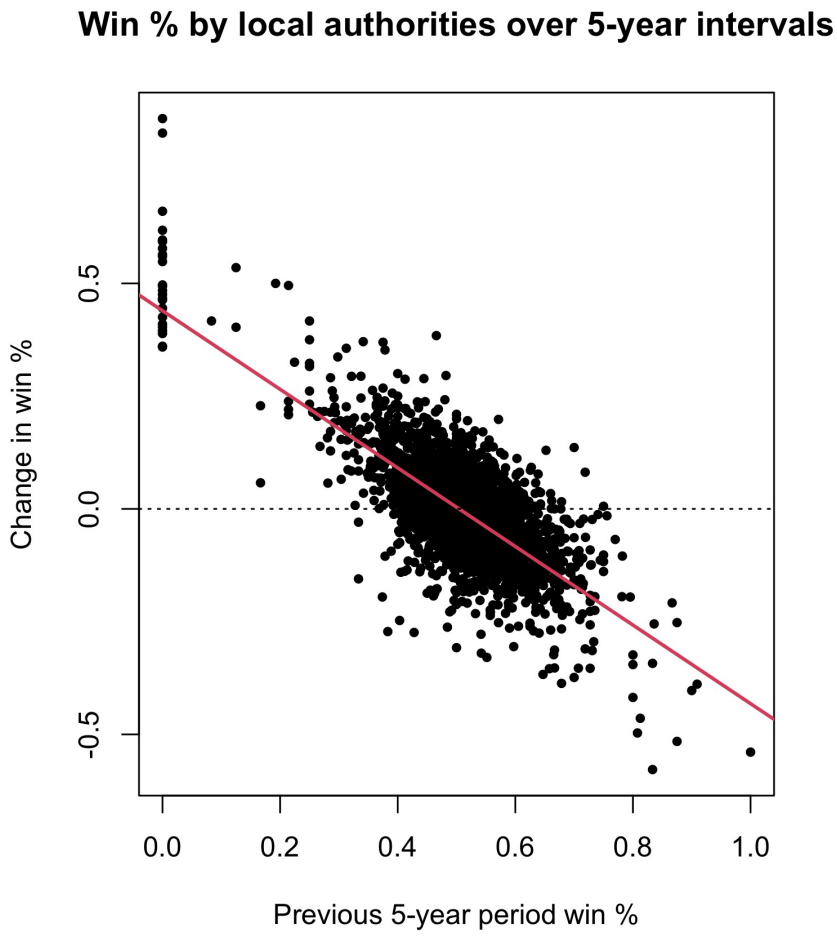
	Dependent variable:													
	outcome													
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)	(14)
	1880-00	1890-10	1900-20	1910-30	1920-40	1930-50	1940-60	1950-70	1960-80	1970-90	1980-00	1990-10	2000-20	2010-30
Elo prediction	0.422***	0.489***	0.558***	0.410***	0.492***	0.529***	0.482***	0.438***	0.487***	0.565***	0.566***	0.616***	0.649***	0.642***
	(0.083)	(0.048)	(0.042)	(0.041)	(0.034)	(0.038)	(0.038)	(0.031)	(0.025)	(0.022)	(0.021)	(0.021)	(0.021)	(0.025)
Full position (H)	−0.005***	−0.001	−0.0001	−0.001***	−0.001***	−0.001	−0.0002	−0.002***	−0.002***	−0.002***	−0.003***	−0.003***	−0.003***	−0.003***
	(0.001)	(0.001)	(0.001)	(0.0004)	(0.0003)	(0.0004)	(0.0004)	(0.0003)	(0.0002)	(0.0002)	(0.0002)	(0.0002)	(0.0002)	(0.0002)
Full position (A)	0.003*	0.002***	0.001**	0.002***	0.001***	0.001	0.001**	0.002***	0.002***	0.002***	0.002***	0.002***	0.002***	0.002***
	(0.001)	(0.001)	(0.001)	(0.0004)	(0.0003)	(0.0004)	(0.0004)	(0.0003)	(0.0002)	(0.0002)	(0.0002)	(0.0002)	(0.0002)	(0.0002)
Population share	0.021	−0.293**	−0.093	−0.014	0.011	−0.055	0.018	0.010	0.037*	−0.030	−0.017	0.186	0.239**	0.082
	(0.204)	(0.119)	(0.120)	(0.081)	(0.028)	(0.070)	(0.084)	(0.047)	(0.022)	(0.024)	(0.107)	(0.119)	(0.116)	(0.138)
Experience share	0.137***	0.067***	0.026**	0.031***	0.028***	0.005	−0.008	0.0004	0.014*	0.028***	0.020***	0.030***	0.024***	0.009
	(0.028)	(0.014)	(0.013)	(0.011)	(0.009)	(0.010)	(0.010)	(0.008)	(0.008)	(0.007)	(0.007)	(0.006)	(0.006)	(0.007)
Observations	5,005	14,947	17,564	24,740	36,443	26,697	28,213	46,523	55,792	64,228	68,827	70,019	70,715	50,768
R^2	0.144	0.095	0.070	0.053	0.052	0.060	0.055	0.050	0.063	0.070	0.069	0.070	0.073	0.076
Adjusted R^2	0.123	0.084	0.062	0.045	0.046	0.054	0.047	0.044	0.057	0.064	0.063	0.064	0.067	0.069
Residual Std. Error	0.390 (df = 4880)	0.392 (df = 14775)	0.394 (df = 17399)	0.393 (df = 24553)	0.393 (df = 36243)	0.397 (df = 26505)	0.404 (df = 27973)	0.402 (df = 46218)	0.394 (df = 55433)	0.397 (df = 63849)	0.404 (df = 68400)	0.408 (df = 69579)	0.412 (df = 70290)	0.414 (df = 50395)
Note:														*p<0.1; **p<0.05; ***p<0.01

Results — convergence

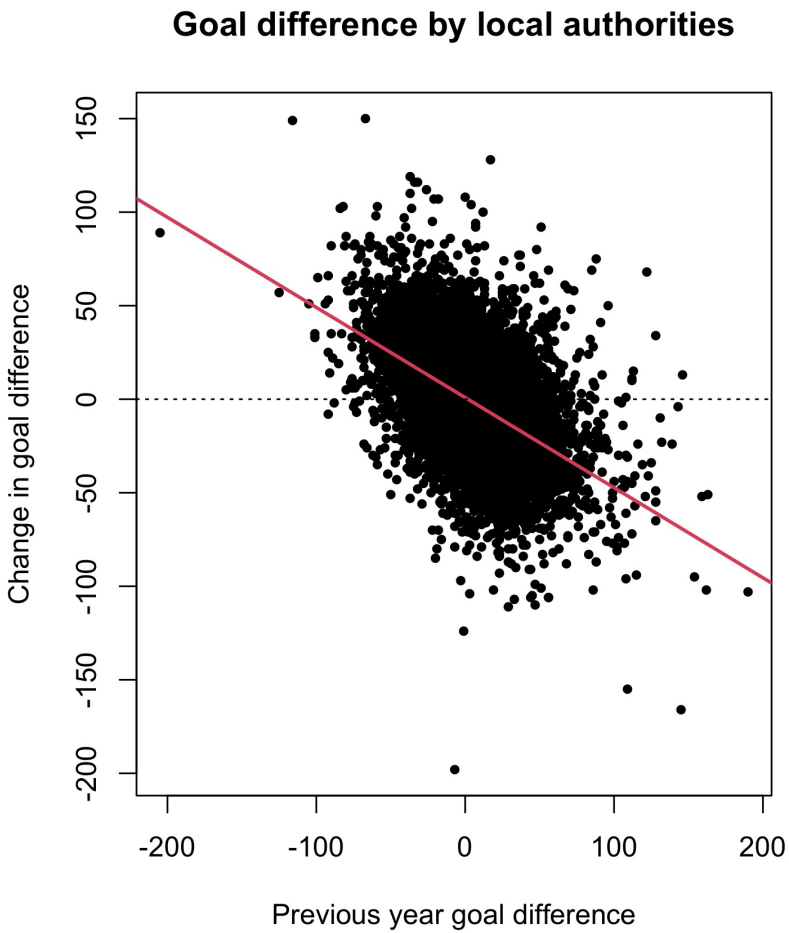
	Dependent variable:			
	Change in win %			
	One-year		Five-year	
	All	$N > 25$	All	$N > 25$
	(1)	(2)	(3)	(4)
Constant	0.366***	0.309***	0.381***	0.437***
	(0.003)	(0.004)	(0.006)	(0.007)
Lagged win %	−0.785***	−0.614***	−0.782***	−0.864***
	(0.006)	(0.007)	(0.013)	(0.014)
Observations	24,538	14,270	5,370	3,396
R^2	0.396	0.324	0.415	0.525
Adjusted R^2	0.396	0.324	0.415	0.525
Residual Std. Error	0.200 (df = 24536)	0.088 (df = 14268)	0.131 (df = 5368)	0.074 (df = 3394)
F Statistic	16,091.440*** (df = 1; 24536)	6,838.662*** (df = 1; 14268)	3,808.647*** (df = 1; 5368)	3,749.210*** (df = 1; 3394)
Note:				*p<0.1; **p<0.05; ***p<0.01

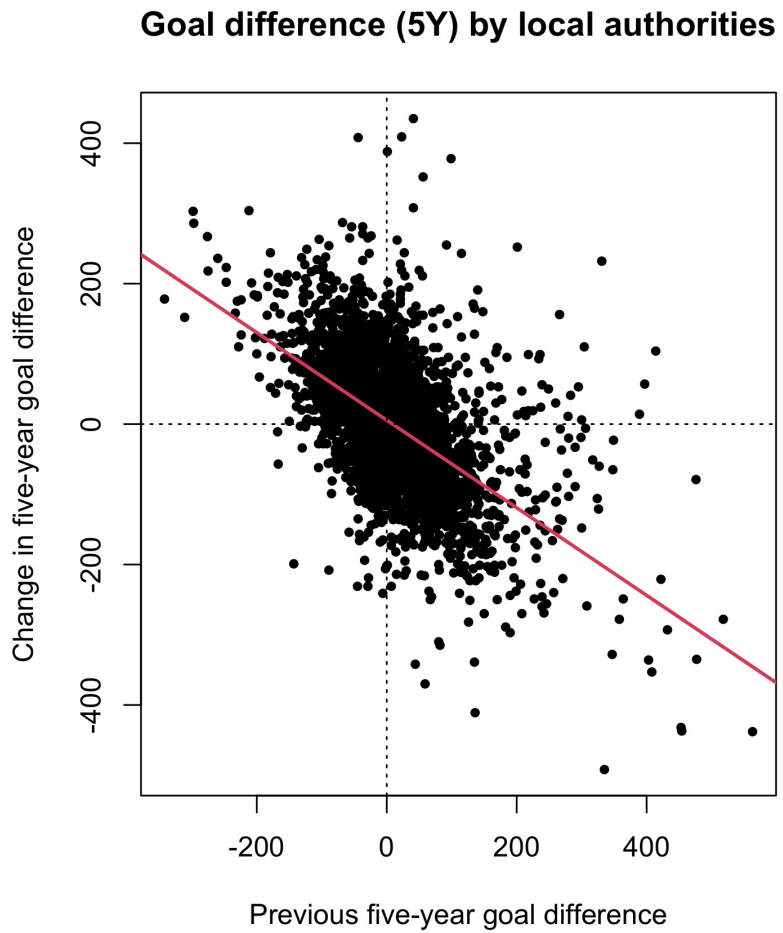
Results — convergence





Results — convergence





Results — convergence

	Dependent variable:			
	Change in goal difference			
	One-year		Five-year	
	All	$N > 25$	All	$N > 25$
	(1)	(2)	(3)	(4)
Constant	0.804***	0.969***	4.570***	5.891***
	(0.113)	(0.188)	(0.870)	(1.370)
Lagged goal difference	−0.486***	−0.465***	−0.608***	−0.590***
	(0.005)	(0.007)	(0.013)	(0.016)
Observations	24,538	14,270	5,370	3,396
R^2	0.243	0.223	0.303	0.281
Adjusted R^2	0.243	0.223	0.303	0.281
Residual Std. Error	17.647 (df = 24536)	22.308 (df = 14268)	63.305 (df = 5368)	78.821 (df = 3394)
F Statistic	7,858.682*** (df = 1; 24536)	4,103.947*** (df = 1; 14268)	2,337.215*** (df = 1; 5368)	1,325.359*** (df = 1; 3394)
Note:				*p<0.1; **p<0.05; ***p<0.01

Results

- Evidence supporting unconditional convergence:
  - Negative relationship between past output level and growth.
- Sigma convergence: Less variability in output? Evidence
- Convergence clubs: Major regions? NUTS1 NUTS2

Conclusions

- Regional football convergence *may* be unconditional:
  - Maybe not dependent on local area population/income.
- In recent times any 'convergence clubs' may be disappearing.
- Good news for football clubs *anywhere*...
  - ...though it may be costly.

Bibliography I

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Economic Models of Growth

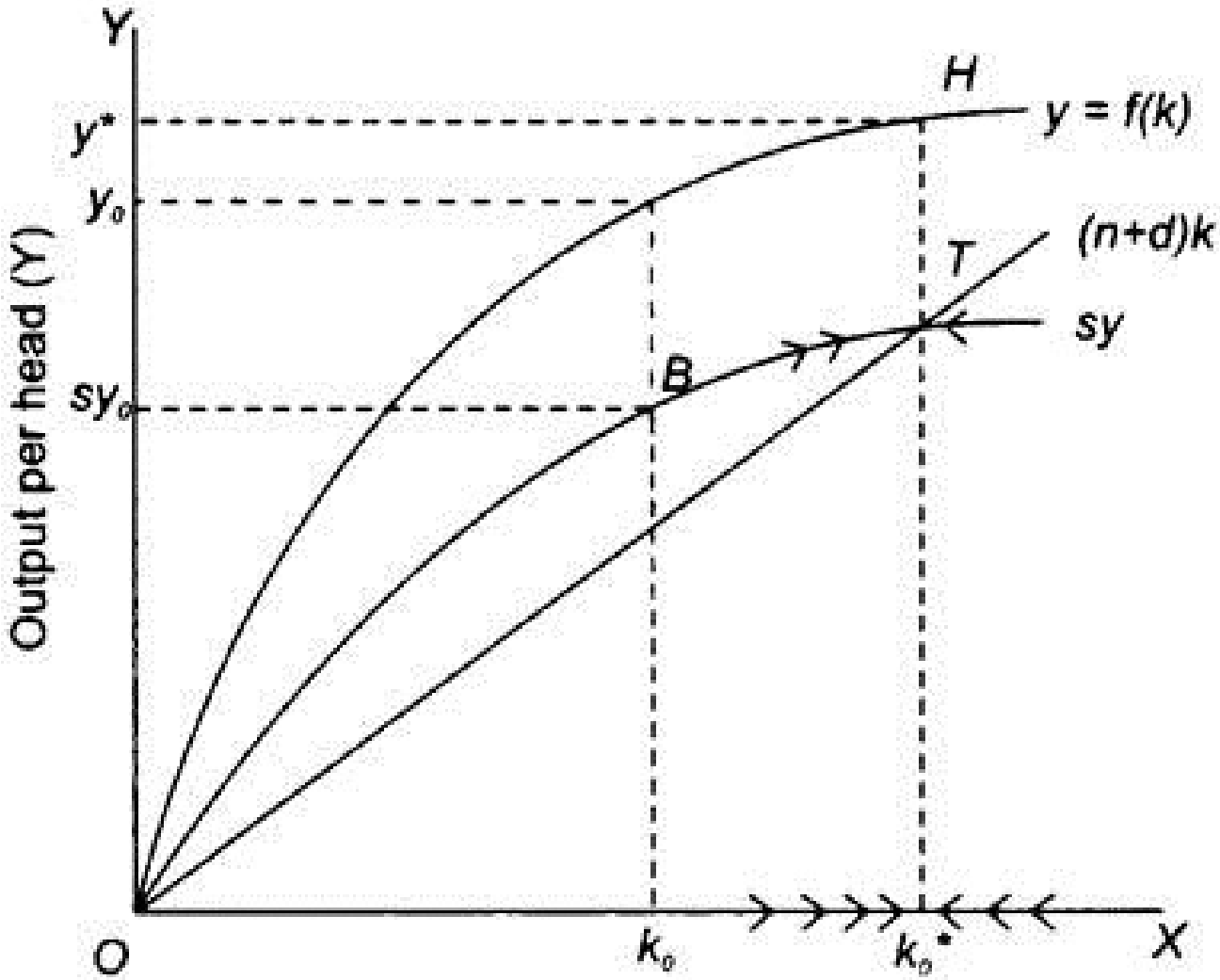


Fig. 14.2. *Neoclassical model: Growth process and steady-state equilibrium*

Method

- Production function:

$$Y = f(A, K, H, L), \tag{5}$$

- Y output produced:
  - Wins (and draws). Weighted by opposition?
- K physical capital: Stadium, training facilities.
- H human capital: Owners, chairmen, managers, players, support staff.
  - What role supporters?
- A technology: Training, health, nutrition, analytics...
  - Adoption of technology needs access to finance/capital?
- Returns to any investment function of local population:
  - Will local economic circumstance affect footballing output?
- Diminishing marginal returns assumed in (5).

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Elo ratings and predictions

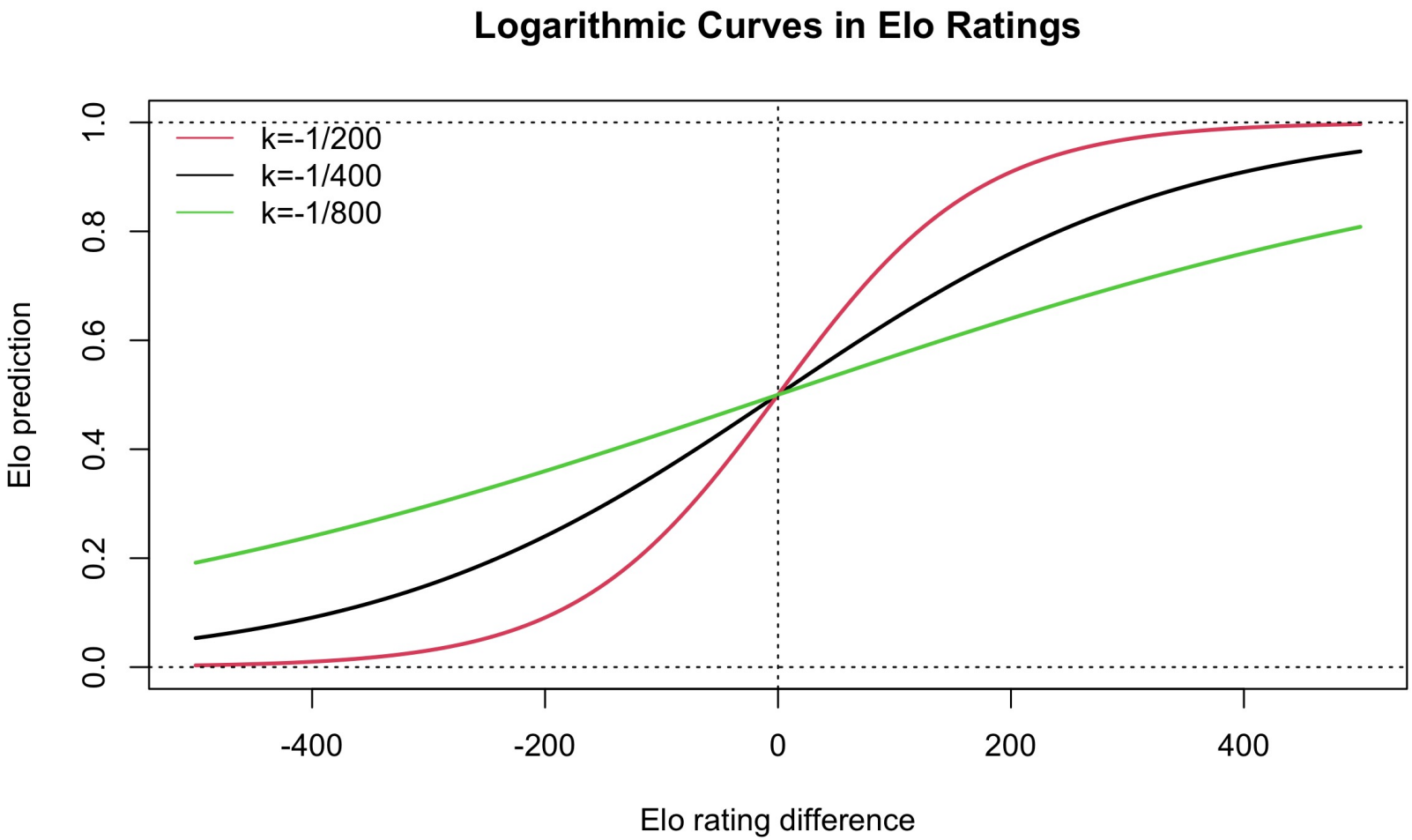
- Elo ratings assign strength rating to each participant.
- For two participants at time t, R_{it}, R_{jt}.
- R_{it}, R_{jt} used to create prediction:

$$elo_{ijt} = \frac{1}{1 + 10^{(R_{jt} - R_{it})/400}}. \tag{6}$$

- By design, $0 \leq elo_{ijt} \leq 1$, for participant j $elo_{jit} = 1 - elo_{ijt}$.
- Outcome Y_1, ratings adjust: $R_{t+1,1} = R_{t,1} + K(Y_1 - X_1)$.

Concept Example Back

Elo ratings and predictions



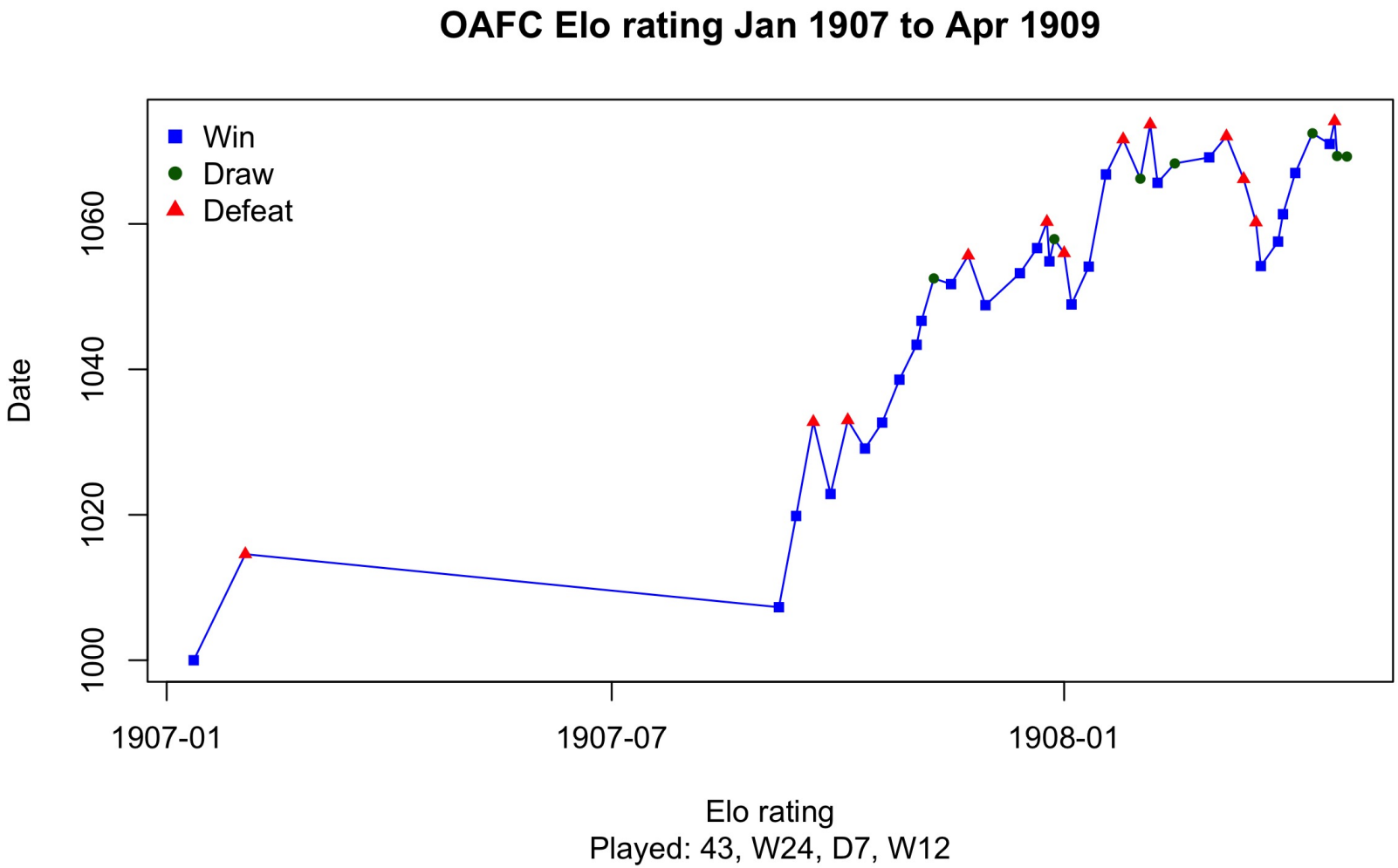
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Elo ratings example: Oldham Athletic

	Home	Goals		Away	Elo ratings			Probabilities		
Date	team	Home	Away	team	Home	Prediction	Away	Home	Draw	Away
1907-01-12	Oldham	5	0	Kidderminster	1000.0	0.51	990.5	50.2	24.6	25.3
1907-02-02	Oldham	0	1	Liverpool	1014.6	0.24	1211.8	24.3	24.3	51.4
1907-09-07	Stoke	1	3	Oldham	1097.7	0.63	1007.3	61.9	20.8	17.3
1907-09-14	Oldham	2	1	West Brom	1019.8	0.35	1124.7	33.9	26.2	39.9
1907-09-21	Bradford	1	0	Oldham	1036.1	0.50	1032.8	49.2	24.8	26.0
1907-09-28	Oldham	3	0	Hull	1022.9	0.49	1028.1	47.9	25.1	27.0
1907-10-05	Derby	1	0	Oldham	1110.8	0.61	1033.0	60.2	21.4	18.4
1907-10-12	Oldham	4	0	Lincoln	1029.1	0.64	926.3	63.5	20.1	16.3
1907-10-19	Fulham	1	2	Oldham	1096.2	0.59	1032.7	58.2	22.2	19.7
1907-10-26	Oldham	1	0	Barnsley	1038.6	0.52	1024.5	50.9	24.4	24.7

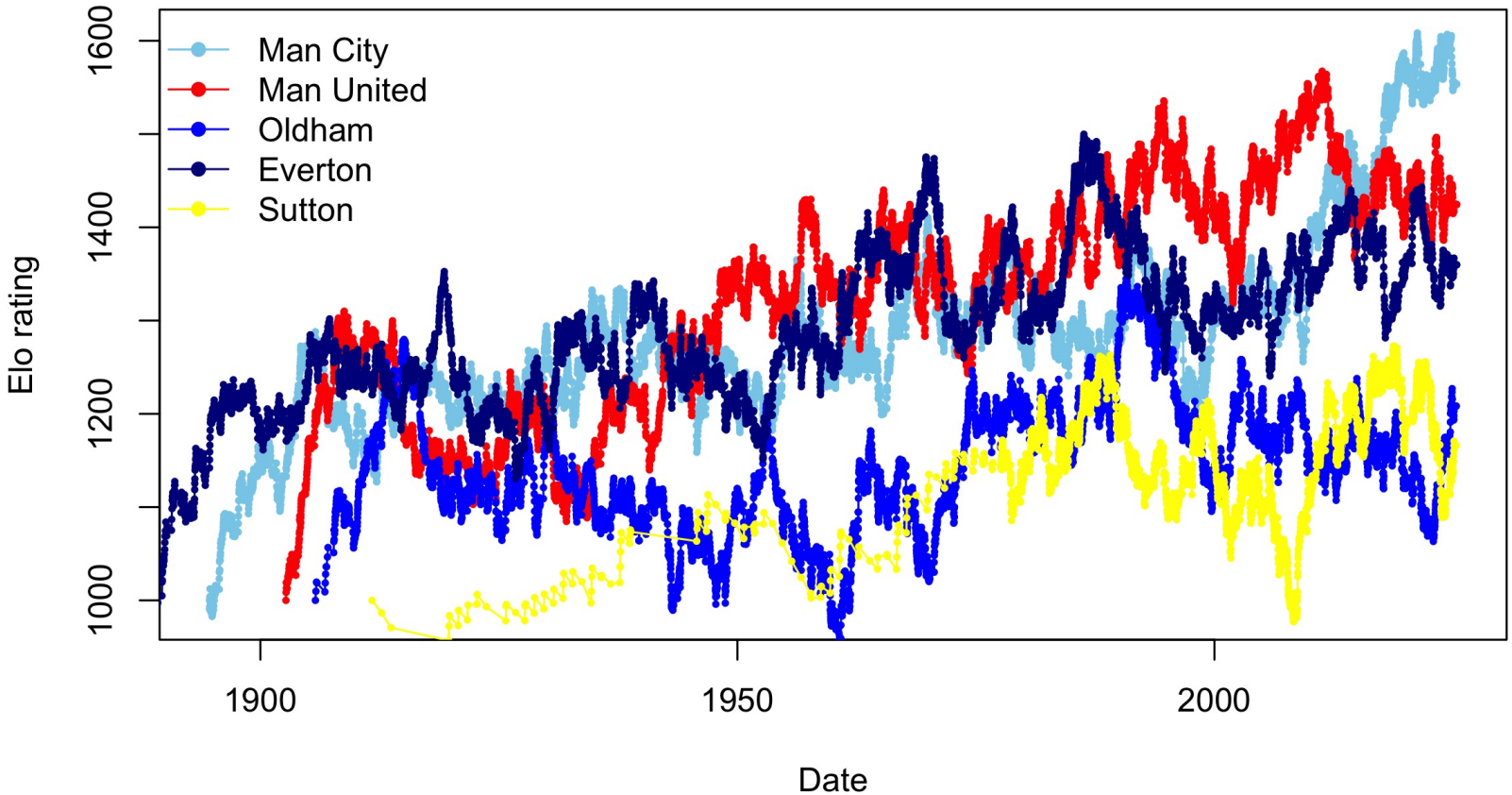
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Elo ratings example: Oldham Athletic



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Elo ratings example: Oldham Athletic



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Elo top 10

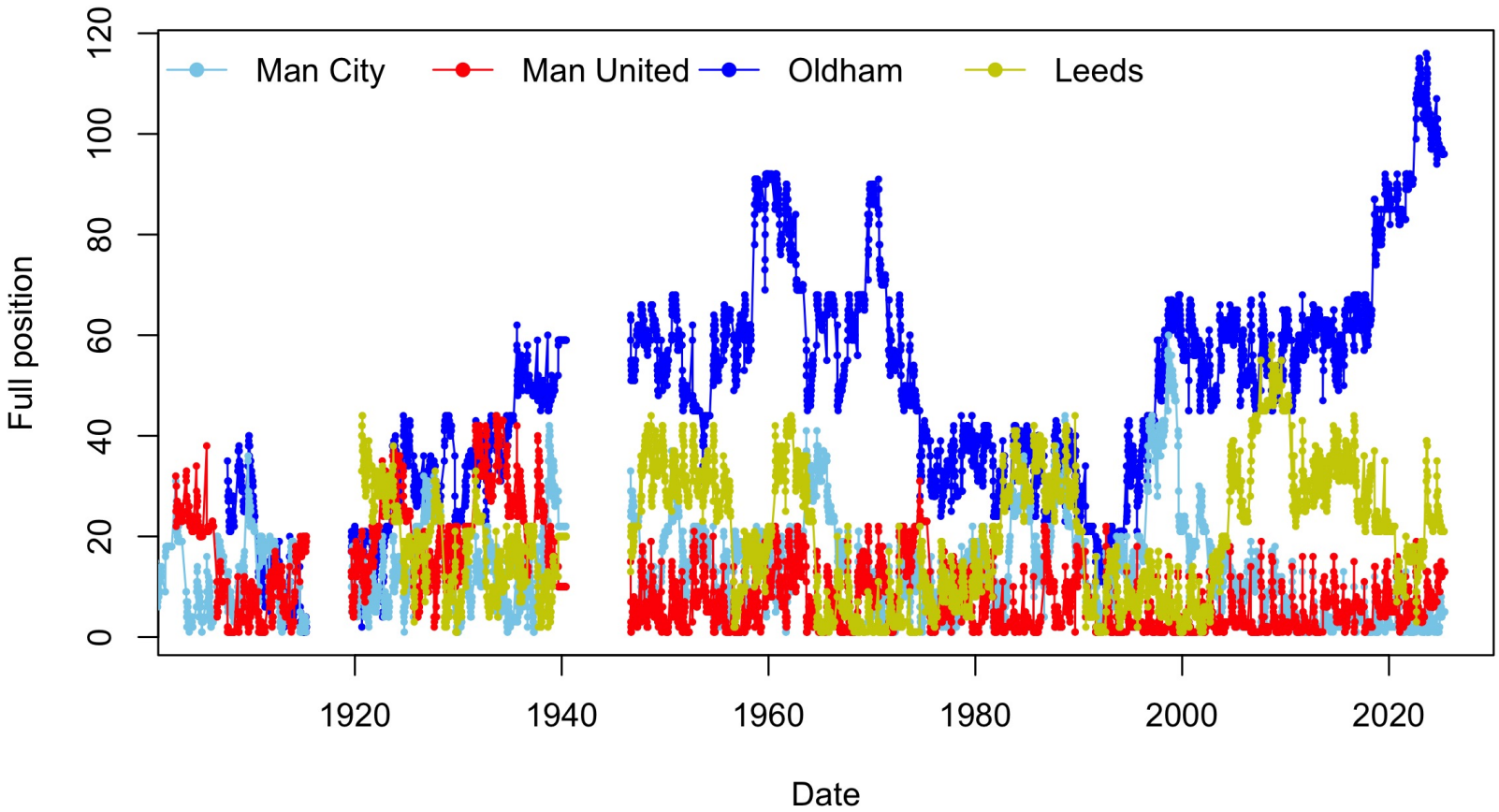
Rank	team	elostrength	division	league position
1	Manchester City	1554	English Premier	5
2	Liverpool	1546	English Premier	1
3	Arsenal	1515	English Premier	2
4	Newcastle United	1481	English Premier	6
5	Chelsea	1452	English Premier	4
6	Brighton and Hove Albion	1429	English Premier	10
7	Manchester United	1424	English Premier	13
8	Tottenham Hotspur	1422	English Premier	14
9	Aston Villa	1412	English Premier	8
10	Fulham	1408	English Premier	9

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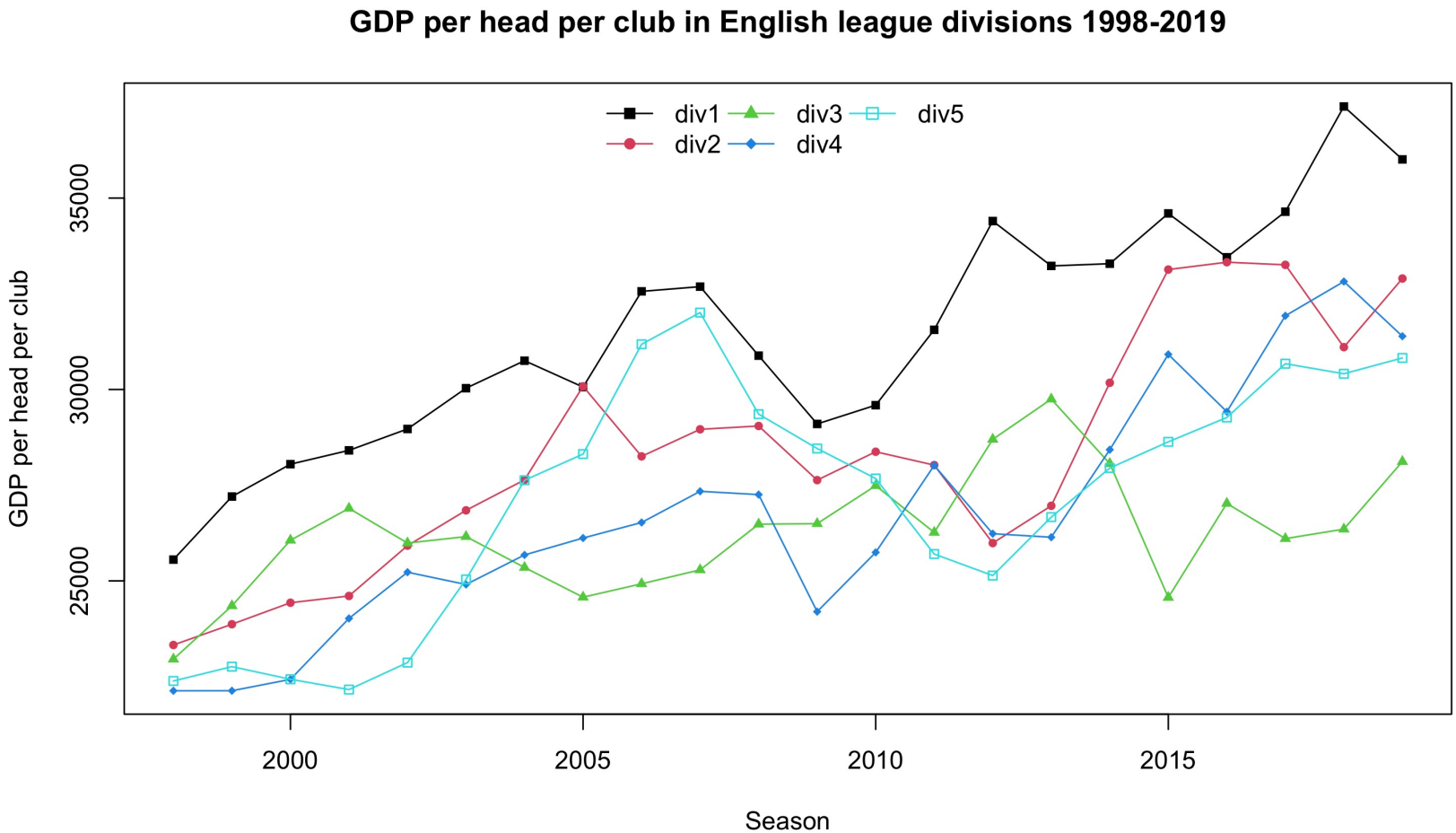
Elo divisional mix

Rank	Team	elostrength	division	league pos
61	Exeter City	1215	Football League One	18
62	Walsall	1215	Football League Two	1
63	Hull City	1214	Football League Championship	20
64	Chorley	1213	Football Conference North	4
65	Doncaster Rovers	1213	Football League Two	5
66	Huddersfield Town	1209	Football League One	6
67	Oldham Athletic	1208	Football Conference	4
68	Cardiff City	1208	Football League Championship	19
69	AFC Wimbledon	1207	Football League Two	4
70	Halifax Town	1207	Football Conference	6

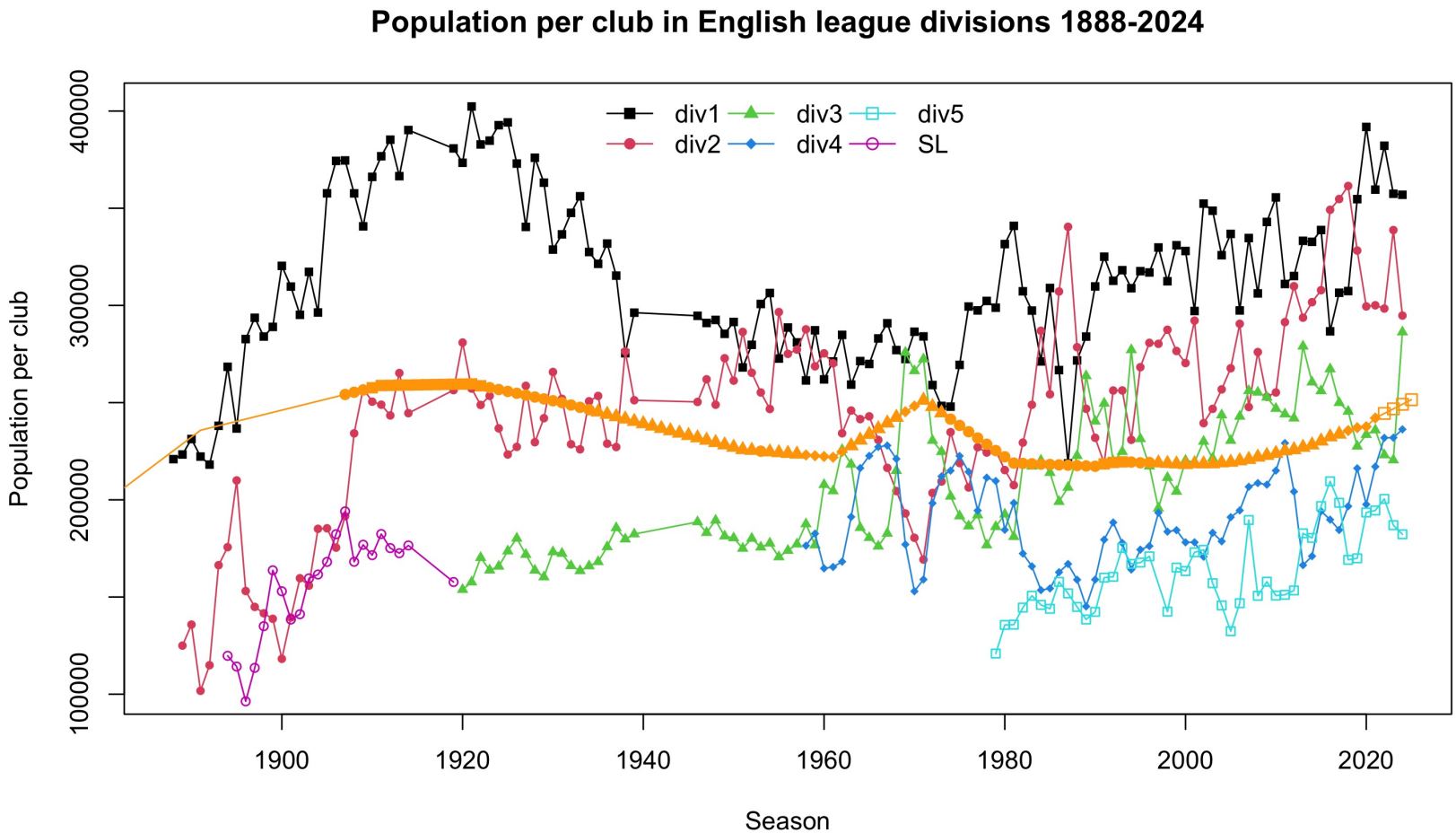
Full league position



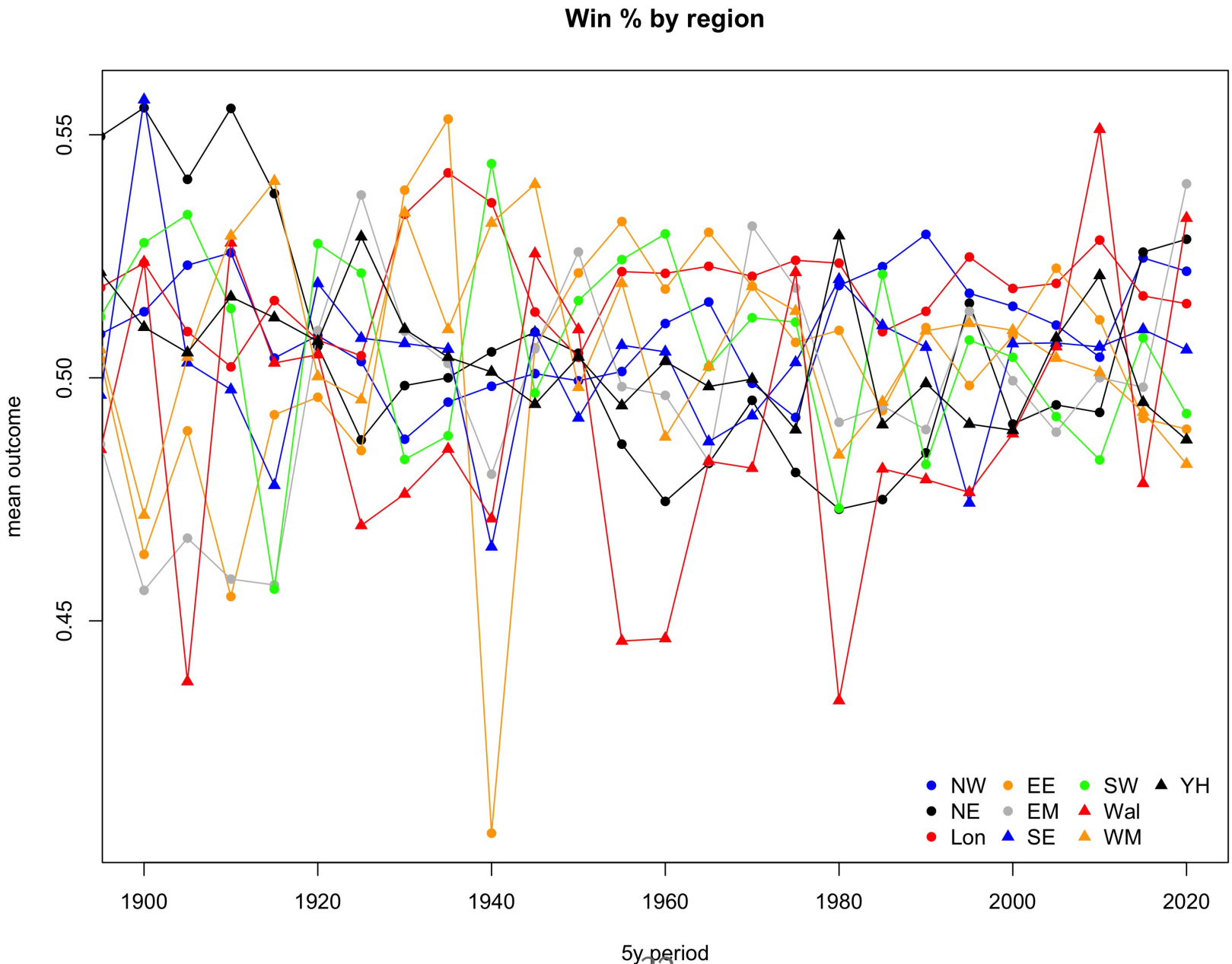
Population per club by division



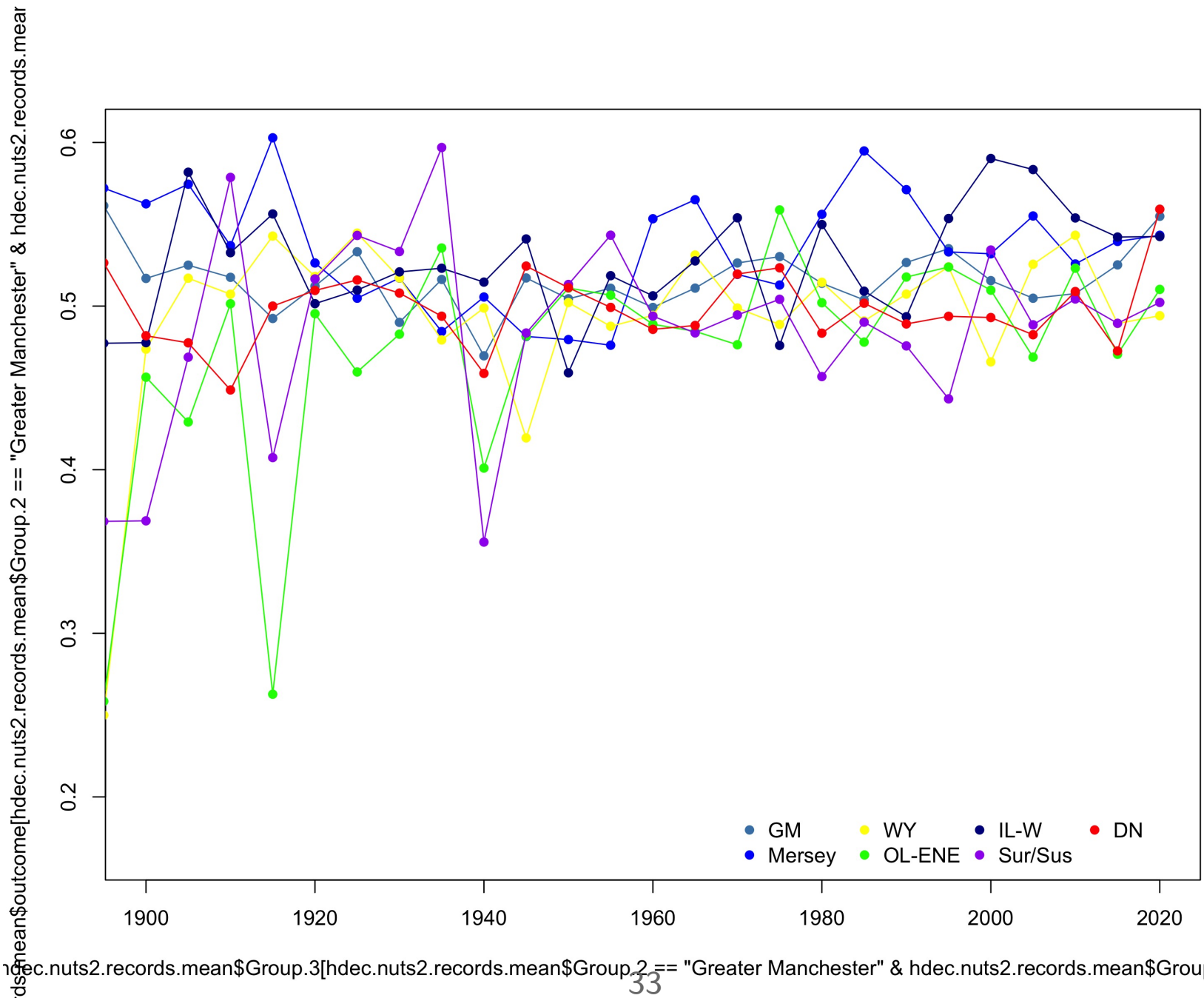
Population per club by division



Win percentage by NUTS1 Region



Win percentage by NUTS2 Region



Sigma Convergence

